ΠΑΝΕΠΙΣΤΗΜΙΟ ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
UNIVERSITY OF WEST ATTICA

DEPARTMENT OF ENGINEERING
INFORMATION AND COMPUTER

SIGNALS AND SYSTEMS

FINAL PROJECT

Name : Athanasiou Vasileios Evangelos
Registration Number: 19390005
Laboratory Department: Department 1 Friday 9:00-11:00
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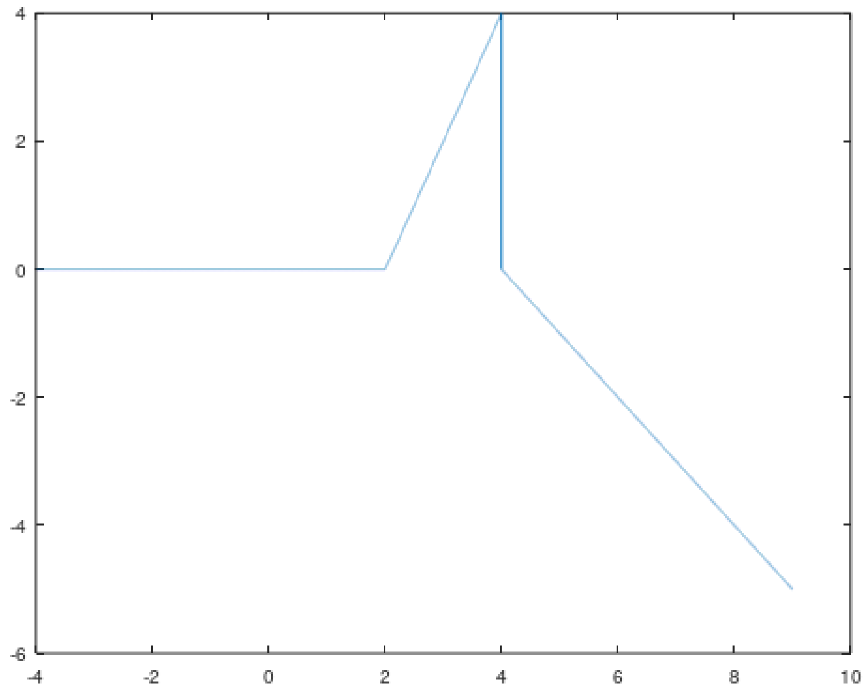
CONTENTS

EXERCISE 1

Commands :

```
step = 0.001;
n1 = -4:step:2-step;
x1 = zeros(size(n1));
n2 = 2:step:4-step;
x2 = 2*n2 - 4;
n3 = 4:step:9;
x3 = 4 - n3;
n = [n1 n2 n3];
x = [x1 x2 x3];
plot( n,x )
```

Graphic representations:



Documentation:

Exercise 1 asks us to design the signal:

$$x[n] = \begin{cases} 0 & \text{if} \quad n < 2 \\ 2n - 4 & \text{if} \quad 2 \le n < 4 \\ 4 - n & \text{if} \quad 4 \le n \end{cases}$$

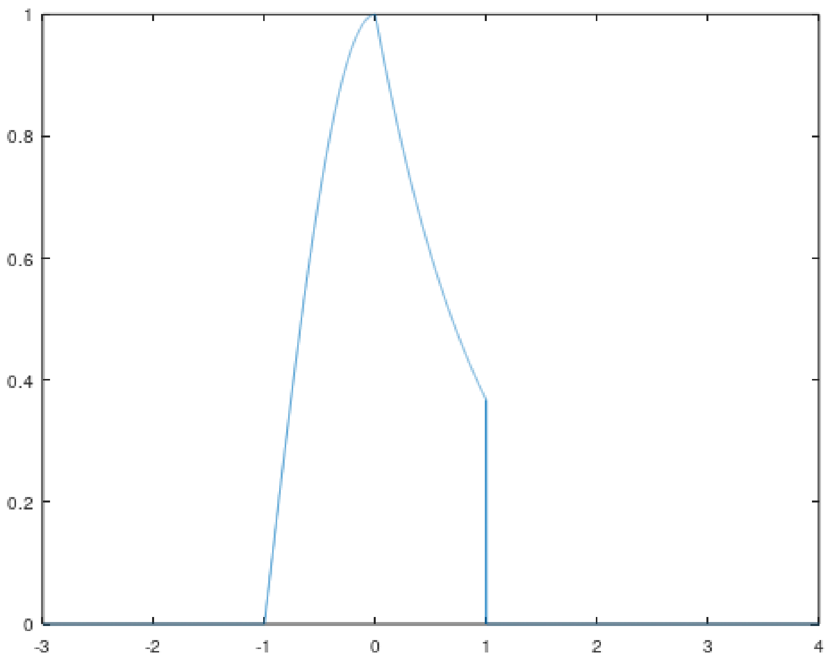
First, we enter the number 0.001 in a " step " variable. Next, we define the first time interval in the vector named " n 1", where it is from -4 to 2-0.001, and in the variable " x 1" we define a dynamic array of zeros, thanks to the function " zeros (size ())' to the size of the vector ' n 1', as the first branch of the complex function is the equation '0'. Then we follow the same procedure to plot the remaining two branches of the function. In detail, we define the second time interval in the vector named " n 2", where it is from 2 to 4-0.001, and in the variable " x 2" we enter the equation of the second branch "2* n 2 – 4". Exactly the same for the third time interval named " n 3", where it is from 4 to 9, and in the variable " x 3" we enter the equation of the third branch "4 - n 3". Finally, in a vector " n " we register the three subvectors " n 1, n 2, n 3" and in a vector " x " the three subvectors " x 1, x 2, x 3" and for designing the signal with function " plot ()" and arguments the vectors " n , x ".

EXERCISE 2

Commands :

```
step = 0.001;
t1 = -3 :step :-1-step;
x1 = zeros(size(t1));
t2 = -1 :step:0 -step;
x2 = cos (pi*t2/2);
t3 = 0 :step:1 -step;
x3 = exp(-t3);
t4 = 1 :step:4 ;
x4 = zeros(size(t4));
t = [t1 t2 t3 t4];
x = [x1 x2 x3 x4];
plot ( t , x )
```

Graphic representations:



Documentation :

H exercise 2 asks us to plot the analog signal:

Για t <= -1, x(t) = 0
Για -1 < t <= 0, x(t) = cos(pi t/2)
Για 0 < t <= 1, x(t) = exp(-t)
Για 1 < t, x(t) = 0

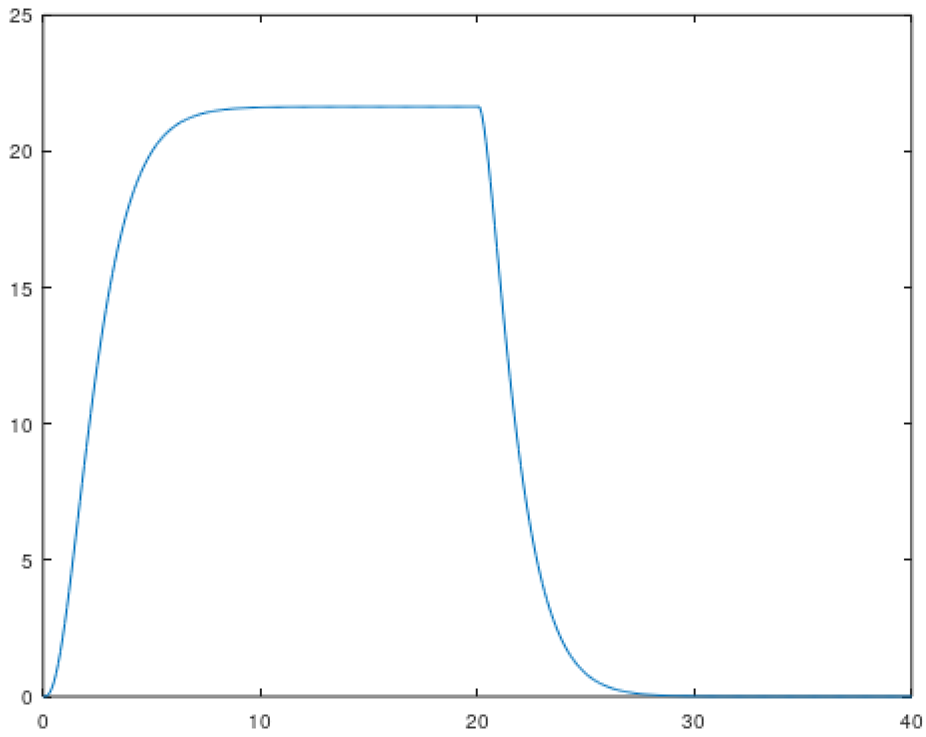
First, we enter
the number 0.001 in a " step " variable. Next, we define the first time interval in the vector named " t 1 ", where it is from -3 to -1-0.001, and in the variable " x 1 " we define a dynamic array of zeros, thanks to the function " zeros (size ())" to the size of the vector " t 1", since the first branch of the complex function is the equation "0". Then we follow the same procedure to plot the remaining three branches of the function. In detail, we define the second time interval in the vector named " t 2", where it is from -1 to 0-0.001, and in the variable " x 2" we enter the equation of the second branch " cos (pi * t 2/2)". Exactly the same for the third time interval named " t 3", where it is from 0 to 1-0.001, and in the variable " x 3" we enter the equation of the third branch "exp(-t3)". For the fourth time interval named " t 4", where it is from 1 to 4, and in the variable " x 4" we define a dynamic matrix with zeros, thanks to the function " zeros (size ())" in the size of the vector " t 4", since the fourth branch of the complex function is the equation "0". Finally, in a vector " t " we register the four subvectors " t 1, t 2, t 3, t 4" and in a vector " x " the four subvectors " x 1, x 2, x 3, x 4" and for plotting the signal with the " plot ()" function and the " t , x " vectors as arguments .

EXERCISE 3

Commands :

```
t = 0:0.1:10;
h = (2*t.* exp( -t) + exp(-2*t) - exp(-3*t)) .*
heavyside (t);
x = (1 - exp( -1.5*t)) .* heavyside (t);
y = conv ( x ,h ).* 1;
plot( 0:0.1:20,y )
```

Graphic representations:



Documentation:

Exercise 3 asks us to calculate and plot the convolution of continuous-time functions:

$$h(t) = \left[2te^{-t} + e^{-2t} - e^{-3t}\right] \cdot u(t)$$

$$x(t) = \left[1 - e^{-1.5t}\right] \cdot u(t)$$

First, we define a time interval in the variable " t " from 0 to 10. Then, we construct the GMA system with impulse response and the input signal with with the help of the " heaviside " function and we register the system in the variable " h " and the signal in the variable " x ". Then, we register in the variable " y " the result of the convolution of the continuous time functions with the help of the " conv " function. Finally, to plot the convolution we used the " plot ()" function with arguments twice the interval of the variable " t ", so that the size of the vectors in the arguments of " plot ()" also match the " y ".

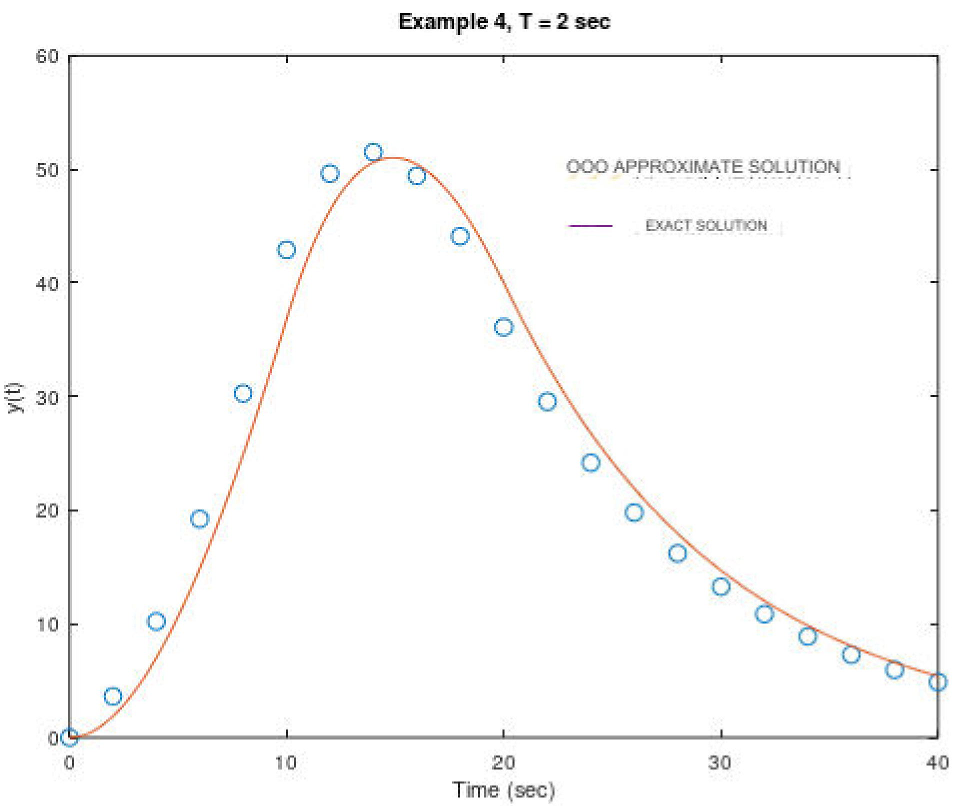
EXERCISE 4

Commands :

```
n=0:20;
kf = 0.1; m = 1; T = 2;
h = (1- exp( - kf *n*T/m))/ kf ;
x = [ ones( 1,5) -ones(1,5) zeros(1,10)];
y = conv ( T* h,x );

t1 = 0:.1:9.9;
t2 = 10:.1:19.9;
t3 = 20:.1:40;
ya = [100*(0.1*t1-1+exp(-0.1*t1)),-100*(0.1*t2-
3+(2*exp(1)-1)...
* exp( -0.1*t2)),100*(1-2*exp(1)+exp(1)^2)*exp(-
0.1*t3)];
t = [t1 ,t2,t3 ];
plot( n* T,y (1:length(n)),' o',t,ya ,'-')
title( 'Example 4, T = 2 sec')
xlabel ( 'Time (sec)' )
ylabel ( 'y(t)' )
hold on
plot([23 24 25],[50 50 50],'o',[23 24 25],[45 45
45]);
text(26,50, 'APPROXIMATE SOLUTION')
text(26,45, 'EXACT SOLUTION')
hold off
pause
```

Graphic representations:



Documentation:

The code is split into two parts. The first one with the help of the " conv " function of " Matlab " calculates the convolution of two continuous time functions, while the second one equally with a more complex structure which is simplified thanks to this function. Essentially, the purpose of both parts of the code is the same, just with a different methodology. So, the purpose of the program is to compare the reliability of the two methodologies and see if they show the same result. Also, the two graphics are designed with a different color and shape so that the difference is clearly visible. As we observe, there is a difference between the two members.